Tushar

Software Engineer
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Summary

Full-Stack Software Engineer with 3+ years of experience in scalable applications, distributed systems, and production GenAI. Proficient in React, Node.js, Python, and FastAPI, with expertise in agentic AI, MCP, microservices, event-driven architecture, and asynchronous workflow orchestration.

Work Experience

Software Engineer Nov 2025 – Present
Evalueserve *Gurgaon, India*

- Orchestrated a **distributed AI workflow platform** processing **2K+ daily uploads**, coordinating **asynchronous jobs**, distributed workers, multi-stage AI pipelines, and report delivery through **Redis** and **Azure Blob Storage**.
- Integrated production **agentic AI workflows** with external tools and services using custom **MCP clients and servers**, enabling tool-driven AI execution within business workflows.
- Introduced resilient execution mechanisms including **Redis-backed checkpointing**, state recovery, **idempotency**, retries, and dead-letter handling to resume interrupted workflows and minimize redundant processing.
- Reduced report generation time by approximately **35%** through **parallel execution** of independent workflow stages; handled external API failures with timeouts, rate-limit controls, bounded concurrency, and failure isolation.
- Shaped the platform's **React** interface for workflow configuration, execution status, and **human-in-the-loop (HITL)** approvals, integrating backend APIs, reusable UI components, and **execution observability** for monitoring and failure investigation.

Full Stack Developer July 2024 - October 2025
Ecompapi *Delhi, India*

- Optimized **backend database performance** through **SQL tuning, Redis caching, and query refactoring**, reducing database latency from **200ms to 30-40ms**.
- Built **asynchronous data processing systems** using **Kafka and TimescaleDB**, enabling real-time analytics and processing **100K+ jobs/day** while moving high-volume processing out of synchronous request paths.
- Developed **HLS video processing** with **RabbitMQ**-based asynchronous segmentation and multi-resolution transcoding, generating adaptive video streams while supporting **video tracking, in-video quizzes, and media locking**.
- **Streamlined platform operations** by developing **dynamic course interfaces and user impersonation tooling**, enabling **50+ support agents** to troubleshoot learner issues ∼**40% faster**.

Software Developer November 2023 - June 2024
Utrade Solutions - Fintech solutions firm enabling smarter trading *Mohali, India*

- Integrated **WhatsApp APIs** into the User Management System, enabling **10K+ real-time order alerts/day**.
- Automated **order and position backups** for **1M+ daily trades**, preventing data loss.

Technical Skills

Languages: Python, TypeScript, JavaScript (ES6+), C++, SQL
Backend: FastAPI, Node.js, Express.js, REST APIs
AI & GenAI: Generative AI, Agentic AI, RAG, MCP, LangChain, pgvector
Databases & Caching: PostgreSQL, MongoDB, TimescaleDB, Redis
Messaging & Systems: Kafka, RabbitMQ, Distributed Systems, Event-Driven Architecture, Async Programming, Linux
Cloud & DevOps: Docker, Kubernetes, Azure (Blob, OpenAI, VMs, Functions) , CI/CD

Projects

Crux | ***C++20, Linux, epoll*** | GitHub

- Built an `epoll`-based **event loop** and **coroutine scheduler** from scratch in **C++20**, enabling **non-blocking TCP/HTTP I/O** without relying on external async frameworks.
- Replaced **thread-per-connection** handling with lightweight **C++20 coroutines**, reducing scheduling and memory overhead while keeping asynchronous request handling readable through `co_await`.

Internal Developer Platform | ***FastAPI, React, Docker, Kubernetes*** | GitHub

- Built a **self-service Internal Developer Platform** using **FastAPI, React, Docker, and Kubernetes**, automating **service scaffolding, containerization, and Kubernetes deployment** through a unified developer portal.
- Engineered **automated deployment workflows** supporting **CI/CD, application scaling, health checks, and rollback**, enabling developers to manage the application lifecycle without manual Kubernetes operations.

Education

Chandigarh College of Engineering and Technology Chandigarh, India
Bachelor of Engineering - Computer Science and Engineering *2019 - 2023*